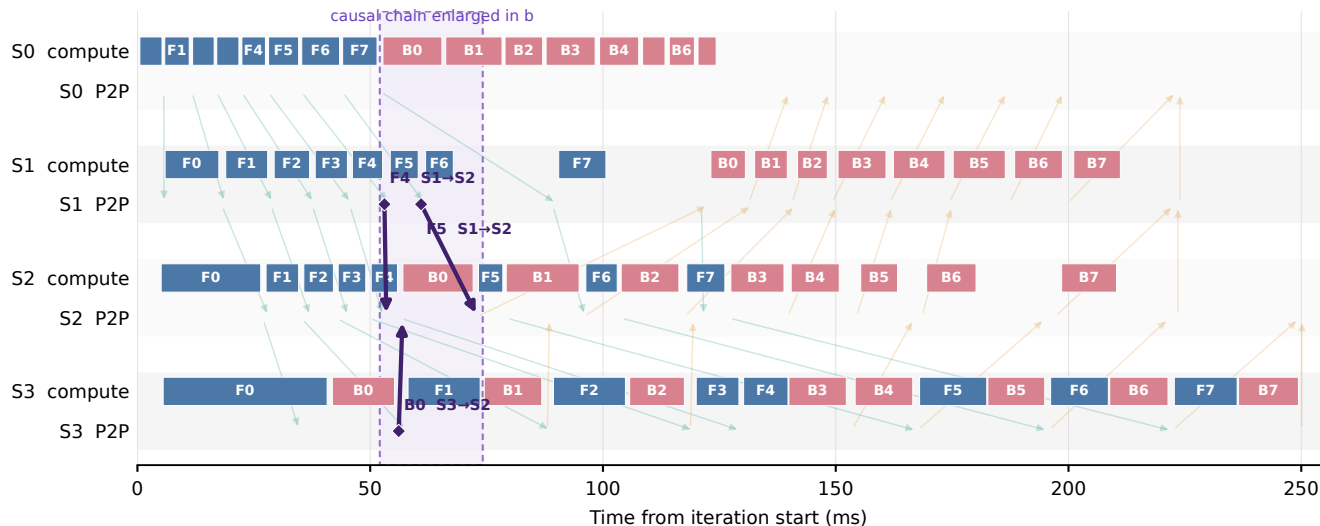
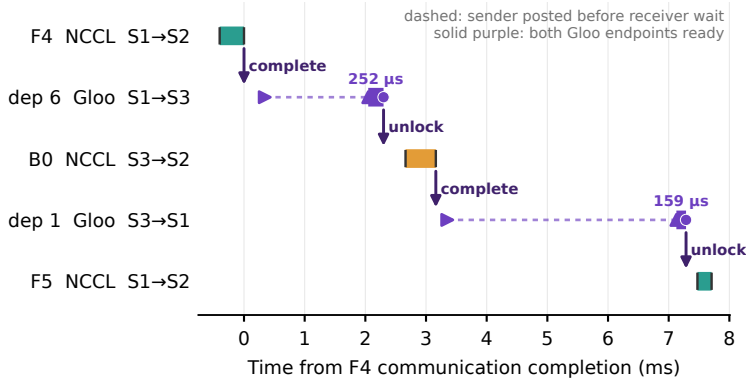


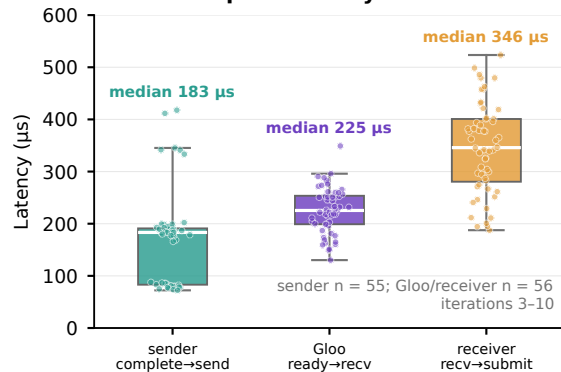
### a C schedule: four-stage execution trace (iteration 5)



### b Measured cross-rank dependency chain



### c Control-path latency



Forward compute Backward compute Forward communication Backward communication Gloo control signal